

FILTER DESIGN TEST CASE

Low-Pass Filter - GPS L5 (1176 MHz) for UWB (Ultra-Wideband) Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Low-Pass Filter filter targeting GPS L5 (1176 MHz) for UWB (Ultra-Wideband) Filter. The filter is designed on Langasite substrate with a target center frequency of 3362.1 MHz and bandwidth of 317.9 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the UWB (Ultra-Wideband) Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Low-Pass Filter
- Frequency Band: GPS L5 (1176 MHz)
- Application: UWB (Ultra-Wideband) Filter
- Substrate: Langasite
- Package: Fan-Out WLP
- Design Tool: Keysight ADS 2024
- Design Center: Helsinki 5G Lab

This specification applies to all variants of the Low-Pass Filter design targeting GPS L5 (1176 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3362.1 MHz
Bandwidth (3 dB):	317.9 MHz
Insertion Loss Target:	≤ 1.5 dB
Return Loss Target:	≥ 12.0 dB
Out-of-Band Rejection:	≥ 27.0 dB
Isolation (Duplexer):	≥ 38.0 dB
Q Factor:	12064
Temperature Coefficient:	-15.2 ppm/°C
Die Size:	1.4 x 2.4 mm
Wafer Size:	8-inch
Number of Resonators:	12
Electrode Material:	Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Keysight ADS 2024 and load the Low-Pass Filter template project.
2. Define the substrate stack: Langasite with specified layer thicknesses.
3. Set design targets: center freq 3362.1 MHz, BW 317.9 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.5 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Fan-Out WLP).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, Langasite).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 10086 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.5 dB
- Return loss in passband shall be >= 12.0 dB
- Out-of-band rejection at +/- 477 MHz offset >= 27.0 dB
- Group delay variation across passband shall be <= 9 ns
- Temperature drift over -40°C to +85°C shall be <= 6.4 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 86%

6. TEST RESULTS

Measurement Date: 2024-06-07
Devices Tested: 82
Insertion Loss (meas): 1.23 dB
Return Loss (meas): 14.1 dB
Rejection (meas): 29.8 dB
Isolation (meas): 43.3 dB
Q Factor (meas): 12064
Group Delay Variation: 1.3 ns
Power Handling: +26 dBm
Die Yield: 86.9%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Low-Pass Filter design demonstrated acceptable performance for GPS L5 (1176 MHz).
- b) Insertion loss met target specification.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Package parasitic effects were consistent with 3D EM model predictions.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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